

The Murty-Simon bound at order 29

Reviewer-v3 computer-assisted candidate proof

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Murty–Simon at n=29: reviewer-v3 candidate proof

11 September 2026. Research directed by Paul Lenz; mathematical development, implementation and internal checking by ChatGPT/Geeps. Hardened after a blind external-assistant red-team.

Status: serious candidate proof; independent mathematical review, external computational reproduction and novelty assessment remain OPEN. This edition reduces the computational dependency surface relative to the 9 September Fan-free v2 proof. It uses the minimal trusted Delta=16 kernel and new hand exclusions for the high Delta=16 upper range.

Historical editions remain preserved unchanged.

1. Statement

Let G be a simple diameter-two edge-critical graph on 29 vertices. The candidate statement is

$$e(G) \leq 210 = \left\lfloor \frac{29^2}{4} \right\rfloor,$$

with equality if and only if

$$G \cong K_{14,15}.$$

A bipartite graph of diameter two is complete bipartite, so the bipartite case is immediate. Hence only the non-bipartite dense case needs analysis.

The published theorem of Dailly, Foucaud and Hansberg gives at most $\text{floor}(29^2/4)-2=208$ edges for a non-bipartite diameter-two-critical graph with a dominating edge, apart from their order-six exception. Thus the dense cases below have no dominating edge.

A universal vertex forces a star and is therefore sparse.

2. Delta=15 witness argument and equality

A **direct witness** is an edge uv whose endpoints have no common neighbour. A **two-step witness** is a nonedge uv whose endpoints have exactly one common neighbour. Every critical

edge is covered by one such witness: deleting the edge produces a pair whose every path of length at most two used that edge.

A direct witness covers one edge; a two-step witness covers at most the two edges of its unique length-two path.

Because there is no dominating edge, every witness pair uv satisfies

$$d(u) + d(v) \leq 28. \quad (2.1)$$

Assume $\Delta=15$. Put

$$\varepsilon_x = 15 - d(x), \quad L = \{x : \varepsilon_x \geq 2\}, \quad O = \{x : \varepsilon_x = 1\}.$$

Let $h=|L|$, $o=|O|$, and

$$T = 29 \cdot 15 - 2e(G).$$

Every witness has deficit sum at least two, so

$$2h + o \leq T. \quad (2.2)$$

Count all edges incident with L directly. An edge entirely outside L is witnessed either by an O - O witness, of capacity at most two, or by a missing L - X pair, of capacity at most one. This gives

$$e(G) \leq \binom{h}{2} + h(29 - h) + o(o - 1). \quad (2.3)$$

No injective map from graph edges to witness pairs is assumed; only witness capacities are used.

At $e(G)=211$, $T=13$. The maximum right sides of (2.3) for $h=0, \dots, 6$ are

156, 138, 127, 123, 126, 136, 153,

all below 211. Thus $\Delta=15$ is impossible at 211 and therefore at every larger edge count as the deficit budget only shrinks.

At $e(G)=210$, $T=15$. The corresponding maxima are

210, 184, 165, 153, 148, 150, 159, 175.

Equality forces $h=0, o=15$. All witnesses then lie inside O . Separating direct O -edges from two-step O -nonedges gives

$$210 \leq e(G[O]) + 2 \left(\binom{15}{2} - e(G[O]) \right) = 210 - e(G[O]),$$

so O is independent. Its 15 vertices each have degree 14 and therefore meet all 14 vertices outside O . Those 210 cross edges exhaust the graph. Hence

$$G = K_{15,14}.$$

3. Self-contained Delta=16 bridge

For Delta=16, put $H = \text{complement}(G)$, choose a minimum-degree vertex v , and set

$$\begin{aligned} A &= N_H(v), & |A| &= a = 12, \\ B &= V(H) \setminus N_H[v], & |B| &= b = 16. \end{aligned}$$

For every missing unordered B-pair choose exactly one cross quasi-edge representative $ui \rightarrow w$, with exception w in B . Call these selected and all other A-B edges residual.

The complete graph-to-model proof is now isolated self-contained in

`project/reviews/n29/2026-09-11-reviewer-v3/GRAPH_TO_MODEL_BRIDGE.md`.

Its necessary consequences include:

$$\begin{aligned} e(F) &= r + t, \\ S &\geq r + 2t, \\ \rho_u &\geq 1 \text{ for every B-source when } t > 0, \\ s_i &\leq \rho_u \text{ at every selected incidence,} \\ r - b &\geq \sum_i s_i(s_i - 1) / (12 - s_i), \\ \sum_i s_i(13 - 2s_i) / (12 - s_i) &\geq 16 + 2t, \\ 2W_h &\leq z_h^2 - z_h + h(h + 1), \\ \delta(C) &\geq 1, \\ d_i &\leq 10, \\ r &\leq 60 - t, \end{aligned}$$

together with the selected-edge, supplement and endpoint-load constraints needed by the late relaxation.

This v3 bridge expands the threshold-capacity proof in full rather than leaving it compressed in the main reviewer surface.

4. Delta=16 at m=210 and 211: minimal trusted kernel

The preferred proof-critical route is the minimal trusted kernel from the restarted hostile audit. It deliberately does **not** depend on the historical projected screen, joint propagator, old shared/typed/endpoint LP stack, or the old pair-capacity support formula.

Its sequence is:

```
graph-to-model bridge
-> exact charging demand domain
-> exact threshold/source-count pruning
-> exact source-capacity Hall dual pruning
-> complete sorted residual-row enumeration
-> monotone supplement-cap refinement
```

-> corrected cumulative-threshold/source-q-flow relaxation
-> exact integer Farkas verification.

4.1 m=211, t=3

The minimal preparation retains 72 demand profiles. Complete residual-row enumeration and necessary Hall/refinement cuts leave 126 rows. Every one of those 126 rows has an exact integer Farkas contradiction in the corrected late model.

retained demands: 72
residual rows: 126
exact rejections: 126
final survivors: 0

4.2 m=210, t=2

The same independently specified kernel gives

retained demands: 367
residual rows: 1467
exact rejections: 1467
final survivors: 0

The late model is the corrected v2 cumulative-threshold/source-q-flow relaxation. The historical v1 model contained a real label-group multiplicity normalization bug and is explicitly quarantined as invalid proof evidence.

Floating-point LP output is never accepted as a proof event. It is used only to propose multipliers. The checker accepts a Farkas contradiction only after exact integer recombination with nonnegative multipliers on inequalities, signed multipliers on equalities, nonnegative resulting variable coefficients, and a strictly negative combined right-hand side.

Therefore Delta=16 is impossible at both 211 and 210 edges.

5. Delta=16 upper range without Fan

For any positive-surplus Delta=16 scope the same bridge applies with
t=m-208.

5.1 m=212, t=4

The exact trusted-kernel extension gives

charging-feasible demand profiles: 2032
threshold rejects: 1706
source-count rejects: 258
exact early Hall-dual rejects: 65
open demand profiles: 3
residual numerical states: 19630
row survivors: 2

exact late Farkas rejections: 2
final survivors: 0

Thus m=212 is excluded.

5.2 m=213 and 214

At m=213 (t=5), the complete charging domain has 586 profiles:

576 threshold rejects,
7 source-count rejects,
3 exact Hall-dual rejects,
0 survivors.

At m=214 (t=6), all 79 charging-feasible profiles are threshold-rejected by exact integer arithmetic.

Thus both scopes are excluded before residual-row enumeration.

5.3 m=215 by hand

For every integer $0 \leq s \leq 11$,

$$\frac{s(13-2s)}{12-s} \leq \frac{5}{2}, \quad (5.1)$$

because

$$\frac{5}{2} - \frac{s(13-2s)}{12-s} = \frac{(s-4)(4s-15)}{2(12-s)} \geq 0. \quad (5.2)$$

Equality occurs only at $s=4$.

At m=215, t=7, the bridge requires the sum of the twelve terms in (5.1) to be at least 30. Since 30 is also their maximum possible total, all twelve demands must equal four. Thus $S=48$.

The charging lower bound gives

$$r - 16 \geq 12 \frac{4 \cdot 3}{8} = 18,$$

so $r \geq 34$. But $S \geq r + 14$ gives $r \leq 34$. Therefore $r=34$.

Let z_4 be the number of residual rows of degree at least four. Residual activity gives

$$34 = r \geq 16 + 3z_4,$$

so $z_4 \leq 6$. Threshold capacity gives

$$96 = 2W_4 \leq z_4^2 - z_4 + 20 \leq 50,$$

contradiction. Hence m=215 is impossible.

5.4 $m \geq 216$ by hand

Summing (5.1) over twelve labels gives a universal upper bound 30. The bridge requires

$$16 + 2t \leq 30,$$

so $t \leq 7$, i.e.

$$m \leq 215.$$

Therefore every $\Delta=16$ scope with $m \geq 216$ is excluded analytically. The historical generic outer scan over $m=216 \dots 232$ is retained only as corroborating evidence and is no longer a logical dependency.

6. $\Delta=17$ by a pointwise charging bound

For $\Delta=17$, $a=11$. The bridge requires

$$17 + 2t \leq \sum_{i=1}^{11} \frac{s_i(12 - 2s_i)}{11 - s_i}.$$

For every integer $0 \leq s \leq 10$,

$$\frac{s(12 - 2s)}{11 - s} \leq \frac{16}{7},$$

because

$$\frac{16}{7} - \frac{s(12 - 2s)}{11 - s} = \frac{2(s - 4)(7s - 22)}{7(11 - s)} \geq 0.$$

Thus the right side is at most $176/7 < 29$. At 210 edges the required left side is already 29, and it only increases with m . Hence $\Delta=17$ is impossible at every edge count relevant here.

7. $\Delta=18$ through 27 by residual h -index

Let h be the largest integer for which at least h residual rows have degree at least h . A label of demand s_i needs s_i distinct selected sources with residual degree at least s_i , so $s_i \leq h$ and hence $S \leq ah$.

Residual activity gives

$$r \geq h^2 + (b - h) = b + h(h - 1).$$

Combining with $S \geq r + 2t$,

$$b + 2t \leq (a + 1)h - h^2 \leq \left\lfloor \frac{(a + 1)^2}{4} \right\rfloor = \left\lfloor \frac{(29 - b)^2}{4} \right\rfloor. \quad (7.1)$$

Every $b=18, \dots, 27$ violates (7.1) already at 210 edges, and increasing m only strengthens the contradiction. $\Delta=28$ is the universal-vertex/star case.

8. Fan-free assembly

We can now exclude every $m \geq 211$ without Fan's theorem.

For $m \geq 212$:

- $\Delta \leq 14$ is impossible by degree sum;
- $\Delta=15$ is excluded by the witness-deficit argument, monotone stronger above 211;
- $\Delta=16$ is excluded by exact finite arithmetic at $m=212, 213, 214$, by the hand threshold contradiction at $m=215$, and by the pointwise $5/2$ cap for every $m \geq 216$;
- $\Delta=17$ is excluded by Section 6;
- $\Delta=18, \dots, 27$ are excluded by Section 7;
- $\Delta=28$ gives a star.

At $m=211$, the same degree split leaves only $\Delta=16$ after the hand cases, and the minimal trusted kernel excludes it.

Therefore

$$e(G) \leq 210.$$

At $m=210$, all non- $\Delta=15$ cases are excluded by the minimal kernel or the hand degree arguments. Section 2 forces the unique equality graph

$$K_{14,15}.$$

This establishes the stated **candidate** order-29 result, subject to the review boundaries below.

9. Independent hostile evidence

A blind external-assistant red-team supplied on 11 September independently attacked the graph-to-model bridge, residual activity, threshold capacity, isolated-C, corrected LP normalization, exact Farkas semantics and the hand assembly, and reported no fatal defect.

It also independently obtained the complete charging-domain census

t=2: 9251
t=3: 4867
t=4: 2032
t=5: 586
t=6: 79
t=7: 1
t=8: 0,

which the project independently regenerated again.

The blind reviewer also reported an exhaustive NetworkX graph-atlas bridge test through order seven. The project independently reproduced it:

21 unlabeled diameter-two-edge-critical isomorphism types,
50 maximum-degree rooted cases,
58 admissible selection configurations,
0 bridge failures.

These small examples have only t in $\{-3, -2, -1, 0\}$, so they do not independently test positive-surplus residual activity or charging. They are evidence for the base quasi-edge/selected/residual bridge only.

The preserved follow-up is

`project/reviews/n29/2026-09-11-blind-external-ai-redteam-v1/FOLLOWUP.md`.

10. Trust boundary

The current highest-value independent review targets are:

1. complement/quasi-edge construction and the one-representative-per-unordered-pair convention;
2. residual activity for $t > 0$;
3. selected-source demand $s_i \leq \rho_u$;
4. threshold capacity;
5. isolated-C;
6. the actual-graph-to-averaged-variable embedding of the corrected late model;
7. exact Farkas checker semantics;
8. independent reproduction of the finite frontiers;
9. the cited Dailly-Foucaud-Hansberg dominating-edge theorem.

A counterexample to any universal bridge lemma overrides every green computation downstream. No unrestricted Murty–Simon theorem, order above the proved fixed-order frontier, external endorsement or novelty determination is asserted here.

Key project dependencies

- Self-contained v3 bridge: `project/reviews/n29/2026-09-11-reviewer-v3/GRAPH_TO_MODEL_BRIDGE.md`.
- Minimal-kernel report: `project/reviews/n29/2026-09-08-redteam-restart-v1/MINIMAL_KERNEL_REPO`.
- Blind external red-team follow-up: `project/reviews/n29/2026-09-11-blind-external-ai-redteam-v1/`.
- Cross-order pointwise caps: `project/research/fan-free-fixed-orders/2026-09-11-pointwise-analyt`.
- Historical Fan-free v2 proof: `project/reviews/n29/2026-09-09-fan-free-v2/PROOF.md`.

Appendix A. Self-contained Delta=16 graph-to-model bridge

11 September 2026. Reviewer-v3 bridge. Research directed by Paul Lenz; mathematical development and checking by ChatGPT/Geeps. Hardened after a blind external-assistant red-team.

Status: candidate mathematics; independent expert review remains open. This version is deliberately self-contained at the main trust boundary. In particular, the threshold-capacity argument is proved here rather than merely cited from a companion note.

1. Setup

Let G be a simple diameter-two edge-critical graph on 29 vertices with $\Delta(G)=16$. Put $H=\text{complement}(G)$. Choose a minimum-degree vertex v of H . Then

$$d_H(v)=29-1-16=12.$$

Set

$$\begin{aligned} A &= N_H(v), & |A| &= a = 12, \\ B &= V(H) \setminus N_H[v], & |B| &= b = 16. \end{aligned}$$

Let $C=H[A]$, and let F be the complement of C on A . Write $d_i=d_F(i)$.

For an edge count $m=e(G)$, define

$$t=m-b(29-b)=m-208.$$

The proof-critical dense scopes are $t=2,3,4$ for $m=210,211,212$; the upper-range hand arguments below also use larger t .

2. Quasi-edges and one representative per missing unordered B-pair

Take a missing pair uw of $H[B]$. Adding uw to H corresponds to deleting the critical edge uw from G . In $H+uw$ a new adjacent total-dominating pair appears.

That pair must use u or w because no other adjacency changed. It cannot be $\{u,w\}$ because both still miss v . Hence, after interchanging u,w if necessary, there is an existing edge ui of H such that

$$N_H(u) \cup N_H(i) = V(H) \setminus \{w\}.$$

The auxiliary i lies in A because the pair must dominate v . Write

$$ui \rightarrow w.$$

For every missing **unordered** pair $\{u,w\}$ in $H[B]$, choose exactly one such cross-edge, after orienting the pair if necessary. Call it **selected**. Every other existing A-B edge is **residual**.

A selected edge determines its B-source u and its unique exception w . Therefore it recovers its indexing missing unordered B-pair. Consequently:

1. two different missing unordered B-pairs cannot choose the same selected cross-edge;
2. opposite orientations of one unordered pair cannot both be designated selected;
3. at a fixed source, selected labels and supplements are distinct.

For u in B , define

ρ_u = residual degree into A,
 q_u = selected outdegree,
 p_u = selected indegree as supplement/exception.

For i in A, define

R_i = residual degree into B,
 x_i = selected degree into B.

Let

$r = \sum_u \rho_u = \sum_i R_i$.

3. Exact ledger and label demand

Selected cross-edges and existing edges of $H[B]$ partition the unordered pairs of B, so

$\#selected + e(H[B]) = C(b, 2)$.

Direct edge counting gives

$$e(F) = r + t, \quad (3.1)$$

$$\sum_i d_i = 2(r + t). \quad (3.2)$$

Minimum degree in H gives $d_H(i) \geq a$ for every i in A. Since

$$d_H(i) = 1 + (a - 1 - d_i) + R_i + x_i,$$

we have

$$x_i \geq d_i - R_i.$$

Define

$$s_i = \max(0, d_i - R_i),$$

$$S = \sum_i s_i.$$

Then

$$x_i \geq s_i. \quad (3.3)$$

Also

$$\begin{aligned}
 S &\geq \sum_i (d_i - R_i) \\
 &= 2(r + t) - r \\
 &= r + 2t.
 \end{aligned} \quad (3.4)$$

Thus a label of demand s_i genuinely requires at least s_i distinct selected source incidences.

4. Pointwise forcing from a selected edge

Fix a selected edge $ui \rightarrow w$.

4.1 $d_i \leq \rho_u + R_i$

Let j be an F-neighbour of i , so ij is absent from $H[A]$. Since $\{u, i\}$ must dominate j , uj is an H-edge.

If uj is residual, charge j to a residual source slot at u . Otherwise uj is selected, say $uj \rightarrow z$. Distinct selected labels at u have distinct supplements, so $z \neq w$. Since $ui \rightarrow w$ must dominate z and uz is missing in $H[B]$, iz is an H-edge.

Moreover iz cannot be selected: i and z both miss the A-vertex j (ij is an F-edge and j is the unique exception of $uj \rightarrow z$). A selected A-B edge has its unique exception in B and therefore must dominate every A-vertex. Hence iz is residual.

Distinct selected j give distinct supplements z , so the non-residual uj cases inject into distinct residual edges at label i . Therefore

$$d_i \leq \rho_u + R_i. \quad (4.1)$$

In particular, at every selected incidence,

$$s_i \leq \rho_u. \quad (4.2)$$

4.2 $d_i \leq \rho_u + \rho_w$

For the same F-neighbour j , if uj is residual charge it to ρ_u . Otherwise write $uj \rightarrow z$. Since $\{u, j\}$ must dominate w and uw is missing, jw is an H-edge.

The edge jw cannot be selected from source w : j and w both miss the A-vertex i (ij is an F-edge and iw is missing because w is the exception of $ui \rightarrow w$). Hence jw is residual. Distinct j give distinct residual edges at w . Therefore

$$d_i \leq \rho_u + \rho_w. \quad (4.3)$$

4.3 Source degree and supplement forcing

Every F-neighbour j of i is a cross-neighbour of u . Source u has exactly $\rho_u + q_u$ cross-neighbours, one of which is i , so

$$d_i \leq \rho_u + q_u - 1. \quad (4.4)$$

For every other selected $uj \rightarrow z$ at u , the pair $\{u, j\}$ must dominate w . Since uw is missing, jw is an H-edge. The $q_u - 1$ other selected labels are distinct, so w has at least $q_u - 1$ cross-neighbours among them. These are partitioned into residual and selected edges from w , hence

$$\rho_w + q_w \geq q_u - 1. \quad (4.5)$$

5. Missing degree in B and endpoint load

Every missing unordered B-pair incident with u is oriented exactly once, either outward from u or inward to u . Hence

$$q_u + p_u = \text{missing degree of } u \text{ in } H[B]. \quad (5.1)$$

Since

$$d_H(u) = \rho_u + (b-1) - (q_u + p_u) \geq a,$$

we obtain

$$p_u \leq \rho_u + (b-a-1) = \rho_u + 3. \quad (5.2)$$

Also

$$\begin{aligned} q_u + p_u &\leq 15, \\ q_u + \rho_u &\leq 12. \end{aligned} \quad (5.3)$$

For selected $ui \rightarrow w$, label i is adjacent in B to:

- u itself;
- the $q_u - 1$ supplements of the other outward selected pairs at u ;
- the p_u sources of selected pairs oriented into u .

These vertices are distinct. An incoming source cannot equal an outgoing supplement because that would orient the same missing unordered B -pair both ways. Therefore

$$R_i + x_i \geq q_u + p_u. \quad (5.4)$$

6. Residual activity for positive surplus

Assume $t > 0$. We prove

$$\rho_u \geq 1 \text{ for every } u \text{ in } B. \quad (6.1)$$

Suppose instead $\rho_u = 0$. Put

$$\begin{aligned} U &= N_A(u), \\ T &= A \setminus U. \end{aligned}$$

Every cross-edge from u is then selected. There is no F -edge between U and T : if $i \in U$, $j \in T$ and $ij \in F$, the selected edge at ui would have to dominate j , forcing $uj \in H$, contradiction.

Now count forced residual edges.

For every F -edge ij inside U , the selected edges from u to i and j have distinct supplements. Cross-domination forces two residual edges, one at each A -endpoint. They are residual because each forced edge has endpoints that jointly miss an A -vertex. The ordered F -edge endpoints inject into these residual cross-edges, giving $2e(F[U])$ distinct residual edges with A -endpoint in U .

Now take an F -edge ij inside T . Adding ij to H creates a quasi-edge with exception i or j . Its auxiliary cannot be v , because the pair with v already dominates the opposite A -endpoint; it cannot be u because u is adjacent to neither endpoint. If its auxiliary z lay in A , domination of u would force $z \in U$. But a quasi-edge with exception in T would then require an absent U - T pair, contradicting $F(U, T) = \emptyset$. Hence the auxiliary lies in B .

That cross quasi-edge has its unique exception in A , so it cannot be one of the selected representatives whose exception lies in B ; it is residual. Its A -endpoint together with the unique exception recovers the original F -edge, so distinct $F[T]$ -edges give distinct residual cross-edges. Their A -endpoints lie in T and are therefore disjoint from the U -family.

Thus

$$\begin{aligned}
r &\geq 2e(F[U]) + e(F[T]) \\
&\geq e(F[U]) + e(F[T]) \\
&= e(F) \\
&= r + t,
\end{aligned}$$

contradicting $t > 0$. Therefore (6.1) holds, and in particular

$$r \geq b = 16. \quad (6.2)$$

7. Charging inequality

For each label i choose exactly s_i of its actual selected incidences, possible by (3.3). By (4.2), a chosen source u for label i satisfies

$$\rho_u \geq s_i.$$

Also $q_u \leq a - \rho_u$. Charge each chosen incidence from source u by

$$(\rho_u - 1) / (a - \rho_u).$$

A chosen source has $q_u \geq 1$, so $\rho_u \leq a - 1$ and the denominator is positive. Source u participates in at most $q_u \leq a - \rho_u$ chosen incidences, so its total charge is at most $\rho_u - 1$. Summing over all B-sources gives total charge at most

$$r - b.$$

The charge function is increasing in integer ρ on $1 \leq \rho \leq a$. A label with demand s_i therefore receives at least

$$s_i(s_i - 1) / (a - s_i).$$

Hence

$$r - b \geq \sum_i s_i(s_i - 1) / (a - s_i). \quad (7.1)$$

Combining (7.1) with $S \geq r + 2t$ gives

$$\sum_i s_i(a + 1 - 2s_i) / (a - s_i) \geq b + 2t. \quad (7.2)$$

At $a = 12, b = 16$,

$$\sum_i s_i(13 - 2s_i) / (12 - s_i) \geq 16 + 2t. \quad (7.3)$$

8. Threshold-capacity lemma — complete proof

Fix an integer $h \geq 1$ and define

$$\begin{aligned}
I_h &= \{i : s_i \geq h\}, \\
W_h &= \sum_{i \in I_h} s_i, \\
Z_h &= \{u : \rho_u \geq h\}, \\
z_h &= |Z_h|.
\end{aligned}$$

Choose s_i actual selected incidences for each $i \in I_h$. By (4.2), every chosen source belongs to Z_h .

Call a selected incidence **heavy** if its label lies in I_h . Let e_{ll_u} be the number of actual heavy selected incidences from source u . Then

$$W_h \leq \sum_{u \in Z_h} \ell_u. \quad (8.1)$$

Let

$$J = \{u \in Z_h : \ell_u \geq h\}, \\ j = |J|.$$

Fix $u \in J$ and a heavy selected edge $u \rightarrow w$. For each other heavy selected label k at u , $\{u, k\}$ must dominate w ; because uw is missing, kw is an H-edge. There are $\ell_u - 1 \geq h$ such distinct heavy labels k .

We claim $w \in Z_h$. If none of those forced edges kw is selected from source w , then all are residual and $\rho_w \geq \ell_u - 1 \geq h$. If at least one forced edge kw is selected from w , then its label k is heavy, so $s_k \geq h$; applying (4.2) to that selected incidence gives $\rho_w \geq s_k \geq h$. Thus every supplement of a heavy arc from J lies in Z_h .

Therefore every heavy selected arc contributed by a source in J uses an unordered B-pair entirely inside Z_h and incident with J . Because selection is injective on unordered B-pairs, the number of such arcs is at most

$$j(z_h - j) + C(j, 2) = j * z_h - j(j+1)/2. \quad (8.2)$$

Every source in $Z_h \setminus J$ contributes at most h heavy arcs by definition of J . Combining with (8.1),

$$W_h \leq (z_h - j)h + j * z_h - j(j+1)/2. \quad (8.3)$$

If $W_h > 0$, some label has demand at least h and therefore needs at least h distinct sources in Z_h ; so $z_h \geq h$. Put

$$q = z_h - h.$$

Now

$$\begin{aligned} [h * z_h + C(q, 2)] - \text{RHS}(8.3) \\ = (q - j)(q - j - 1)/2 \\ \geq 0, \end{aligned} \quad (8.4)$$

because $q - j$ is an integer and the product of two consecutive integers is nonnegative. Thus

$$W_h \leq h * z_h + C(z_h - h, 2),$$

or equivalently

$$2W_h \leq z_h^2 - z_h + h(h+1). \quad (8.5)$$

This is a necessary capacity bound only; no sufficiency is claimed.

Two useful consequences are immediate. If $H_0 = \max_i s_i > 0$, then

$$z_{H_0} \geq H_0. \quad (8.6)$$

Also, residual activity gives every B-source baseline degree at least one, so for $h \geq 2$,

$$r \geq b + z_h(h-1). \quad (8.7)$$

Hence any upper bound $r \leq r_{\max}$ implies

$$z_h \leq \text{floor}((r_{\max} - b)/(h-1)). \quad (8.8)$$

9. Excluding an isolated vertex of C

Suppose x is isolated in $C=H[A]$. Put $X=A\setminus\{x\}$.

Because x is C -isolated,

$$e(C)=e(C[X]).$$

From the exact ledger $e(C)+r=C(a,2)-t$, the number P_0 of missing H -edges inside X is

$$\begin{aligned} P_0 &= C(a-1,2) - e(C[X]) \\ &= r - (a-1-t). \end{aligned} \tag{9.1}$$

Take a missing pair ij inside X . Adding ij to H creates a new adjacent total-dominating pair using i or j . It cannot be $\{i,j\}$ because both miss x . Suppose it is $iz \rightarrow j$. The auxiliary z cannot be v : v already neighbours j , so adding ij does not newly make $\{i,v\}$ total dominating. It cannot lie in A : every A -vertex other than x misses x or, if $z=x$, ix is not an edge. Hence $z \in B$.

Thus every missing X -pair gives a cross quasi-edge with auxiliary in B and exception in A . Such an edge cannot be selected, because selected representatives have their unique exception in B . Hence it is residual. Distinct missing X -pairs give distinct residual cross-edges because a cross quasi-edge has a unique exception. Let P be this family. Then

$$|P|=P_0=r-(a-1-t). \tag{9.2}$$

Let Z be the set of B -endpoints used by P . For $z \in Z$, choose $iz \rightarrow j$ from P . The pair $\{i,z\}$ must dominate x , and i misses x , so xz is an H -edge. This edge is residual: if xz were selected for a missing B -pair, its selected pair would have to dominate every A -vertex, but $iz \rightarrow j$ implies z misses j while x is isolated in C and also misses j . Thus xz is residual. It lies outside P because its A -endpoint is x .

For each $z \in B \setminus Z$, residual activity supplies a residual edge incident with z ; it is outside P because no edge of P has B -endpoint z . Therefore every one of the b vertices of B supplies a distinct residual edge outside P . Hence

$$r \geq |P| + b.$$

Using (9.2),

$$b \leq a-1-t. \tag{9.3}$$

At $a=12, b=16$, (9.3) is impossible for every positive t of interest. Therefore

$$\delta(C) \geq 1. \tag{9.4}$$

Consequently

$$\begin{aligned} d_i &\leq 10, \\ e(C) &\geq 6, \\ r &\leq C(12,2) - t - 6 = 60 - t. \end{aligned} \tag{9.5}$$

10. The exact bridge passed to the finite kernel

Every actual $n=29$, $\Delta=16$ graph in a positive-surplus scope induces data satisfying:

```

a=12, b=16;
1<=rho_u<=12;
0<=d_i<=10;
r=sum rho=sum R;
e(F)=r+t;
s_i=max(0,d_i-R_i);
S>=r+2t;
q_u+rho_u<=12;
p_u<=rho_u+3;
q_u+p_u<=15;
for selected ui->w:
    d_i<=rho_u+R_i,
    d_i<=rho_u+rho_w,
    d_i<=rho_u+q_u-1,
    rho_w+q_w>=q_u-1,
    R_i+x_i>=q_u+p_u;
charging inequality (7.3);
threshold-capacity inequalities (8.5)-(8.8);
r<=60-t.

```

The finite kernel deliberately enumerates a **superset** of graph-realizable arithmetic states satisfying necessary consequences of this bridge. Eliminating the relaxation is therefore safe provided every pruning rule is necessary and every terminal contradiction is checked exactly.

11. Hand cap for the upper range

For every integer $0 \leq s \leq 11$,

$$s(13-2s)/(12-s) \leq 5/2, \quad (11.1)$$

because

$$\begin{aligned} & 5/2 - s(13-2s)/(12-s) \\ &= (s-4)(4s-15)/(2(12-s)) \geq 0. \end{aligned} \quad (11.2)$$

Equality occurs only at $s=4$. With twelve labels, (7.3) gives

$$16+2t \leq 30,$$

so

$$\begin{aligned} t &\leq 7, \\ m &\leq 215. \end{aligned} \quad (11.3)$$

Thus no $\Delta=16$ graph exists at $m \geq 216$.

At $m=215$, $t=7$, equality in (11.3) forces every demand to equal four. Hence $S=48$. Equation (7.1) gives

$$r \geq 16+12*(4*3/8)=34,$$

while $S \geq r+14$ gives $r \leq 34$, so $r=34$. From residual activity,

$$34 \geq 16+3z_4,$$

hence $z_4 \leq 6$. But threshold capacity gives

$$96 = 2W_4 \leq z_4^2 - z_4 + 20 \leq 50,$$

contradiction. Therefore $m=215$ is also excluded by hand.

The only upper-range $\Delta=16$ scopes above 211 that still require finite arithmetic are consequently

$$m=212, 213, 214.$$

12. Review priorities

The highest-value independent attacks remain:

1. the quasi-edge construction and one-representative-per-unordered-pair convention;
2. the residual/non-selected status of the forced cross-edges in Section 4;
3. residual activity in Section 6;
4. the charging budget in Section 7;
5. threshold capacity in Section 8;
6. the isolated-C injection in Section 9;
7. the graph-to-averaged-LP embedding used after this bridge;
8. exact certificate semantics.

A single counterexample to a universal bridge lemma overrides every green computation downstream. This remains candidate mathematics until independent expert review is completed.